Report task 1

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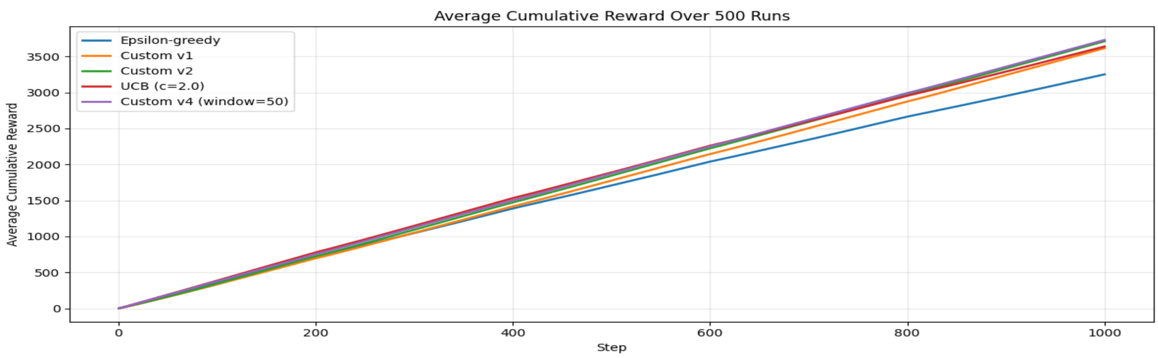
Part 1, analyzing the problem.

Our approach when trying to solve this modified version of the MAB-problem was to first understand what has changed, and then what one had to incorporate to account for the modified problem. The modified version of the problem introduces the possibility of building tolerance for the different medicines, which is incorporated into the code by randomly changing the expected means of the different medicines at some step t. The implication is that whilst one could expect the reward distribution to be stationary before, it is now dynamic and can change over time. Our initial thoughts at this point of the analysis were that compared to previous methods, we needed some sort of change/mechanism to pick up when/if the effectiveness of our medicines (reward distribution) had changed or not.

Part 2, solution algorithm and results comparison.

Our first attempted algorithms were based on landed on epsilon-greedy, where we simply replaced the incremental mean update with: *estimated_rewards[action] += (reward - estimated_rewards[action]) / alpha.* The idea here was that a constant alpha would work better with non-stationary rewards. Next, we attempted to implement a trigger mechanism to detect if the reward distribution changed. This was done by comparing a long moving average against a shorter one and then seeing if this difference exceeded a threshold (chosen as k *std(long_ma) where k was 1.5). This barely yielded any improvement. The final change for our solution was to instead base it on the UCB algorithm, except instead of using all previous rewards to compute action values and confidence bounds, we used a fixed sliding with the 50 most recent observations, which was inspired by our previous iterations.



The results we obtained were reasonable improvements compared to normal epsilon greedy, and marginal improvement compared to normal UCB, with our earlier iterations having roughly equal score as UCB, and our final solution marginally edging on avg. over 500 runs. (Looked at avg. cumulative reward over 500 runs of 1000 eps (steps) to evaluate our solution).